Taller. Preposocesamiento de base datos

Integrantes:

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```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Importación de csv desde github

```
ds=pd.read_csv('https://raw.githubusercontent.com/mateo2408/Taller.-Preprocesamiento-de-base-datos/refs/
```

Varifacar contenido del archivo

```
ds.head()
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	NaN	1.0	3.0	170.0	288.0	0.0	0.0	159.0	0.0	0.2	NaN	NaN	3.0	0.0
1	NaN	1.0	2.0	150.0	168.0	0.0	1.0	174.0	0.0	NaN	2.0	0.0	2.0	1.0
2	45.0	1.0	0.0	115.0	260.0	0.0	0.0	185.0	0.0	0.0	NaN	0.0	2.0	1.0
3	NaN	0.0	2.0	NaN	265.0	1.0	0.0	130.0	0.0	0.0	2.0	1.0	2.0	1.0
4	71.0	0.0	1.0	160.0	302.0	0.0	1.0	162.0	0.0	0.4	2.0	2.0	2.0	1.0

Next steps: Generate code with ds | New interactive sheet

```
var = ds.iloc[:, 2:14].values
print(var)
```

```
[[  3. 170. 288. ...  nan   3.   0.]
 [  2. 150. 168. ...   0.   2.   1.]
 [  0. 115. 260. ...   0.   2.   1.]
 ...
 [  3. 160. 234. ...   1.   2.   1.]
 [  0. 120. 249. ...   0.   3.   0.]
 [  1. 140. 195. ...   2.   2.   1.]]
```

```
ds.describe()
```

	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	
count	298.000000	297.000000	297.000000	297.000000	296.000000	297.000000	297.000000	296.000000	296.000000	295.
mean	1.098994	133.759548	247.074074	0.158249	0.523649	149.367003	0.329966	1.050338	1.402027	0.
std	1.362628	25.223517	52.874097	0.365590	0.520217	23.015491	0.470994	1.176983	0.613848	1.
min	0.000000	94.000000	126.000000	0.000000	0.000000	71.000000	0.000000	0.000000	0.000000	0.
25%	0.000000	120.000000	211.000000	0.000000	0.000000	133.000000	0.000000	0.000000	1.000000	0.
50%	1.000000	130.000000	243.000000	0.000000	1.000000	152.000000	0.000000	0.800000	1.000000	0.
75%	2.000000	140.000000	276.000000	0.000000	1.000000	166.000000	1.000000	1.600000	2.000000	1.
max	7.250026	310.528572	564.000000	1.000000	2.000000	202.000000	1.000000	6.200000	2.000000	4.

```
ds.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 312 entries, 0 to 311
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   age       288 non-null    object 
 1   sex       297 non-null    object 
 2   cp        298 non-null    float64
 3   trestbps  297 non-null    float64
 4   chol      297 non-null    float64
 5   fbs       297 non-null    float64
 6   restecg   296 non-null    float64
 7   thalach   297 non-null    float64
 8   exang     297 non-null    float64
 9   oldpeak   296 non-null    float64
 10  slope     296 non-null    float64
 11  ca        295 non-null    float64
 12  thal      295 non-null    float64
 13  target    297 non-null    float64
dtypes: float64(12), object(2)
memory usage: 34.3+ KB
```

Reemplazo de valores

```
from sklearn.impute import SimpleImputer
imputer = SimpleImputer(missing_values=np.nan, strategy='mean')
imputer.fit(var)
var=imputer.transform(var)
```

```
print(var)
```

```
[[  3.         170.         288.         ...   0.72881356   3.
    0.        ]
 [  2.         150.         168.         ...   0.           2.
    1.        ]
 [  0.         115.         260.         ...   0.           2.
    1.        ]
 ...
 [  3.         160.         234.         ...   1.           2.
    1.        ]
 [  0.         120.         249.         ...   0.           3.
    0.        ]
 [  1.         140.         195.         ...   2.           2.
    1.        ]]
```

Se reemplazan los valores con registro NaN se cambian por valores de media y moda

```
columnas_media = ['trestbps', 'chol', 'thalach', 'oldpeak']
columnas_moda = ['cp', 'fbs', 'restecg', 'exang', 'slope', 'ca', 'thal', 'target']

for col in columnas_media:
    if col in ds.columns:
        mean_val = ds[col].mean()
        ds[col] = ds[col].fillna(mean_val)

for col in columnas_moda:
    if col in ds.columns:
        mode_val = ds[col].mode()[0]
        ds[col] = ds[col].fillna(mode_val)
```

Transformar la columna 0 age de object a float64

```
ds['age'] = pd.to_numeric(ds['age'], errors='coerce')
print(ds['age'].dtype)
print(ds['age'].head())
```

```
float64
0     NaN
1     NaN
2    45.0
3     NaN
4    71.0
Name: age, dtype: float64
```

Transformar la comulna 1 sex de object a float64

```
ds['sex'] = pd.to_numeric(ds['sex'], errors='coerce')
print(ds['sex'].dtype)
print(ds['sex'].head())
```

```
float64
0    1.0
1    1.0
2    1.0
3    0.0
4    0.0
Name: sex, dtype: float64
```

Nueva descripción de datros despues de la transformación

```
ds.describe()
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	ol
count	312.000000	312.000000	312.000000	312.000000	312.000000	312.000000	312.000000	312.000000	312.000000	312.
mean	54.256967	0.708333	1.049680	133.759548	247.074074	0.150641	0.548077	149.367003	0.314103	1.
std	15.943238	0.455260	1.350964	24.607716	51.583241	0.358273	0.517472	22.453596	0.464903	1.
min	-67.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000	71.000000	0.000000	0.
25%	48.750000	0.000000	0.000000	120.000000	212.000000	0.000000	0.000000	135.500000	0.000000	0.
50%	54.256967	1.000000	1.000000	130.000000	245.000000	0.000000	1.000000	151.500000	0.000000	0.
75%	60.000000	1.000000	2.000000	140.000000	274.250000	0.000000	1.000000	165.000000	1.000000	1.
max	109.448965	1.000000	7.250026	310.528572	564.000000	1.000000	2.000000	202.000000	1.000000	6.

```
ds.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 312 entries, 0 to 311
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   age       312 non-null    float64
 1   sex       312 non-null    float64
 2   cp        312 non-null    float64
 3   trestbps  312 non-null    float64
 4   chol      312 non-null    float64
 5   fbs       312 non-null    float64
 6   restecg   312 non-null    float64
 7   thalach   312 non-null    float64
 8   exang     312 non-null    float64
 9   oldpeak   312 non-null    float64
 10  slope     312 non-null    float64
 11  ca        312 non-null    float64
 12  thal      312 non-null    float64
 13  target    312 non-null    float64
dtypes: float64(14)
memory usage: 34.3 KB
```

Limpieza de datos en age

```
age_mean = ds['age'].mean()
ds['age'] = ds['age'].fillna(age_mean)
```

Limpieza de datos en sex

Filtrar la columna sex para que solo permita valores de 0 o 1

Cualquier valor negativo o fuera de este rango se reemplaza por NaN

```
ds.loc[~ds['sex'].isin([0.0, 1.0]), 'sex'] = np.nan

sex_mode = ds['sex'].mode()[0]
ds['sex'] = ds['sex'].fillna(sex_mode)
```

```
ds.head()
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	54.256967	1.0	3.0	170.000000	288.0	0.0	0.0	159.0	0.0	0.200000	2.0	0.0	3.0	0.0
1	54.256967	1.0	2.0	150.000000	168.0	0.0	1.0	174.0	0.0	1.050338	2.0	0.0	2.0	1.0
2	45.000000	1.0	0.0	115.000000	260.0	0.0	0.0	185.0	0.0	0.000000	2.0	0.0	2.0	1.0
3	54.256967	0.0	2.0	133.759548	265.0	1.0	0.0	130.0	0.0	0.000000	2.0	1.0	2.0	1.0
4	71.000000	0.0	1.0	160.000000	302.0	0.0	1.0	162.0	0.0	0.400000	2.0	2.0	2.0	1.0

Next steps: Generate code with ds | New interactive sheet

Mostrar el dataset completamente limpio

```
ds.head(200)
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	targe
0	54.256967	1.0	3.000000	170.000000	288.0	0.0	0.0	159.000000	0.0	0.200000	2.0	0.0	3.0	0.
1	54.256967	1.0	2.000000	150.000000	168.0	0.0	1.0	174.000000	0.0	1.050338	2.0	0.0	2.0	1.
2	45.000000	1.0	0.000000	115.000000	260.0	0.0	0.0	185.000000	0.0	0.000000	2.0	0.0	2.0	1.
3	54.256967	0.0	2.000000	133.759548	265.0	1.0	0.0	130.000000	0.0	0.000000	2.0	1.0	2.0	1.
4	71.000000	0.0	1.000000	160.000000	302.0	0.0	1.0	162.000000	0.0	0.400000	2.0	2.0	2.0	1.
...	...	...	...	...	...	...	...	...	...	...	...	...	...	.
195	54.000000	1.0	0.000000	120.000000	188.0	0.0	1.0	113.000000	0.0	1.400000	1.0	1.0	3.0	0.
196	49.000000	1.0	2.000000	118.000000	149.0	0.0	1.0	126.000000	0.0	0.800000	2.0	3.0	2.0	0.
197	42.000000	1.0	7.250026	120.000000	240.0	1.0	1.0	194.000000	0.0	0.800000	0.0	0.0	3.0	1.
198	48.000000	1.0	1.000000	110.000000	229.0	0.0	1.0	168.000000	0.0	1.000000	0.0	0.0	2.0	0.
199	43.000000	1.0	2.000000	130.000000	315.0	0.0	1.0	149.367003	0.0	1.900000	2.0	1.0	2.0	1.

200 rows × 14 columns

Next steps: Generate code with ds | New interactive sheet